# DSA 8070: Intro to Applied Multivariate Analysis

## Course Information

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# Welcome

## How this course is organized

Learn concepts → Practice in `R` → Apply to projects

### Key idea

The course emphasizes interpretation, computation, and responsible application of multivariate methods

# About the Instructor

- Associate Professor of Applied Statistics and Data Science
- Born in Laramie, Wyoming, and grew up in Taiwan
- B.S. in Mechanical Engineering; transitioned to Statistics in graduate school
- Ph.D. in Statistics, Purdue University, 2017

## Why I mention this

Multivariate analysis is used across engineering, science, business, and data science; the course is designed with that interdisciplinary perspective in mind

# How to Reach Me

- **Email:** wkhuang@clemson.edu
- **Office:** O-221 Martin Hall
- **Weekly office hour:** Wednesday, 8:00–9:00 PM ET, via Zoom

## Communication

Email is the best channel for individual questions. **Please include DSA 8070 in the subject line of your email**. Use Canvas announcements and modules for course-wide information and materials

# Course Structure

**Weekly learning**

- Lecture slides / videos
- R-based practice
- Weekly R labs

**Major applications**

- Project I
- Project II
- Project III

**Key idea**

The projects are designed to integrate statistical reasoning, computation, interpretation, and communication.

# Important Logistics

- There will be three projects:
  - Project I: Sep. 24
  - Project II: Nov. 5
  - Project III: Dec. 3
- Weekly R Labs are submitted through Canvas by 11:59 PM ET on the due date.
- The lowest R Lab grade will be dropped.
- No lectures during Thanksgiving week (Nov. 23–27).

Always use Canvas for the most current deadlines and announcements.

# Course Materials on Canvas

**Course information**

- Syllabus
- Announcements
- Weekly modules

**Learning materials**

- Slides and videos
- R Labs and projects
- Data sets and supporting files

`whitneyhuang83.github.io`

# Reference Books

- Daniel Zelterman (2015), *Applied Multivariate Statistics with R*.
- Alan Izenman (2008), *Modern Multivariate Statistical Techniques: Regression, Classification, and Manifold Learning*.
- Alvin Rencher and William Christensen (2012), *Methods of Multivariate Analysis*, 3rd ed.
- Richard Johnson and Dean Wichern (2008), *Applied Multivariate Statistical Analysis*, 6th ed.

## How to use the books

No single text is followed page-by-page. Readings are used as references that complement the lecture notes and R examples.

# Evaluation

## Weights

| | |
|---|---|
| Industrial Spotlight | 5% |
| R Labs | 20% |
| Project I | 25% |
| Project II | 25% |
| Project III | 25% |

## Letter grades

| | |
|---|---|
| 90.00–100 | A |
| 88.00–89.99 | A- |
| 85.00–87.99 | B+ |
| 80.00–84.99 | B |
| 78.00–79.99 | B- |
| 75.00–77.99 | C+ |
| 70.00–74.99 | C |
| 68.00–69.99 | C- |
| $\leq 67.99$ | F |

# Computing: R and RStudio

We will use `R` and `RStudio` throughout the course.

- Free and open-source tools for statistical computing
- Used to reproduce lecture examples
- Used in R Labs and project analyses

`www.r-project.org` `RStudio`

## Quick check

Before Week 1, can you open RStudio and run a simple command such as `mean(c(1,2,3))`?

# Topics and Course Progression

| Week | Topic |
|---|---|
| 1 | Introduction and multivariate data exploration |
| 2 | A short review of matrix algebra |
| 3 | Multivariate normal distributions and copula models |
| 4 | Inference and comparison of mean vectors |
| 5–6 | Multivariate regression |
| 7 | Inference for covariance matrices |
| 8 | Principal component analysis |
| 9 | Factor analysis |
| 10 | Canonical correlation analysis |
| 11 | Discrimination and classification |
| 12 | Cluster analysis |
| 13 | Multidimensional scaling and distance embedding |
| 14 | Manifold learning and nonlinear embedding |
| 15 | Thanksgiving week – no class |
| 16 | Review |

## Key idea

The course moves from data representation and probability models to inference, prediction, dimension reduction, and unsupervised learning.